

CPE-301.1001

Group Number: 56

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05/10/26

FINAL PROJECT (part 3)

*the Final Project is a solo project, the sole member is Uvensis Martinez

1. Project Motivation/Introduction:

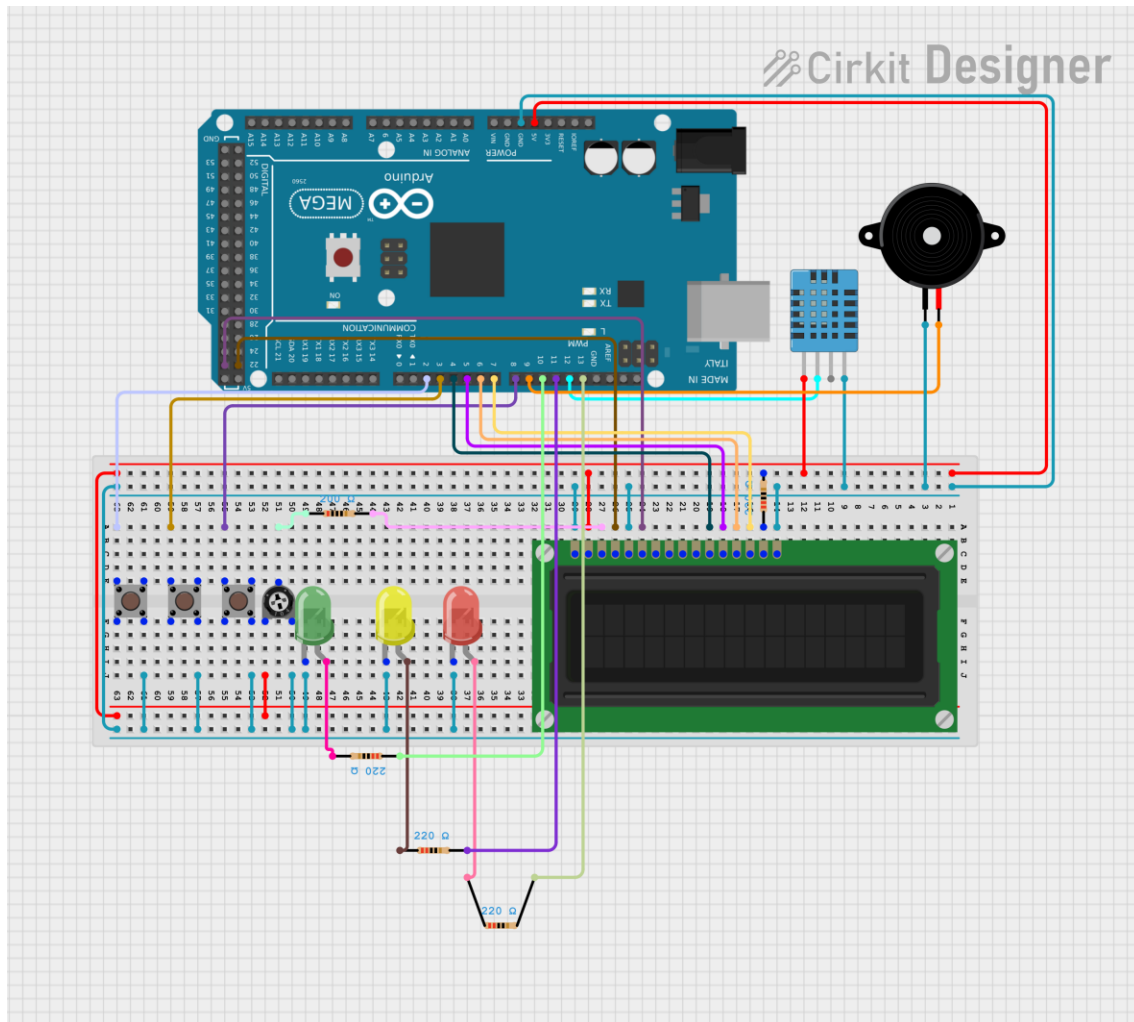
I planned to build a temperature and humidity manager for a terrarium. To ensure that the terrarium is at an acceptable temperature and humidity. If it's not, then it will alert the owner and will alarm in case the terrarium reaches a dangerous temperature for the animal. Owners of lizards and other animals have thermometers in the terrarium to ensure it's at an acceptable temperature. But it is up to the owner to check out the thermometer and humidity regardless of what it senses. Unlike regular sensors, the device will have the ability to alert the owner of any serious issue. It will sound like an alarm with a buzzer and a red LED to alert the owner to fix the issue as soon as possible. This is to ensure the safety of the animal. This device is built for owners of a terrarium.

2. System overview:

The device uses multiple components centered on the temperature and humidity sensor. Once powered and turned off, the device will constantly monitor the temperature and humidity levels of the terrarium of a Bearded Dragon. The Sensor will receive information and send the results to the control system; there it reacts based on the results. It also sends the results to be displayed on the LCD screen. It has 3 reactions to the information (each will send what to display to the LCD)

- 1st (idle) The levels for both temperature and humidity are at an acceptable level and will turn on
- 2nd (Danger) one or both are at a dangerous level which will cause harm to the Bearded Dragon, It will activate the Red LED and activate a Buzzer for an audible and visual alarm to alert the user to act immediately. Each reaction will be displayed on the LCD Screen. warnings will be displayed in case of any issues.
- 3rd ERROR state, it cannot read anything or the device doesn't work properly.

2.1 System schematic Diagram:



**circuit connections are accurate but loosened for readability. The actual device is more compact*

2.2 Component List:

- LCD 1602 Module
- DHT11 Temperature and Humidity Module
- RED LED
- Yellow LED
- Green LED
- Button (4x)
- Passive Button
- Potentiometer 10k
- Mega Controller Board

- Jumper Wires
- Female to male Dupont Wire
- Resistors (220)
- 9V1A Adapter

3. System Behavior:

When the system is ON, it will constantly monitor and check the temperature and humidity of where the sensor is placed; it reads the data and sends it to be displayed through the LCD.

When the System is OFF, the LEDS, and LCD will shut down; the system does not read the temperature and humidity. The system will not function until the user presses a button, turning on the whole system. In case of any errors (loose connections or the system isn't reading anything) the system will enter ERROR mode, it will check if it can work properly, otherwise it will stay in error mode

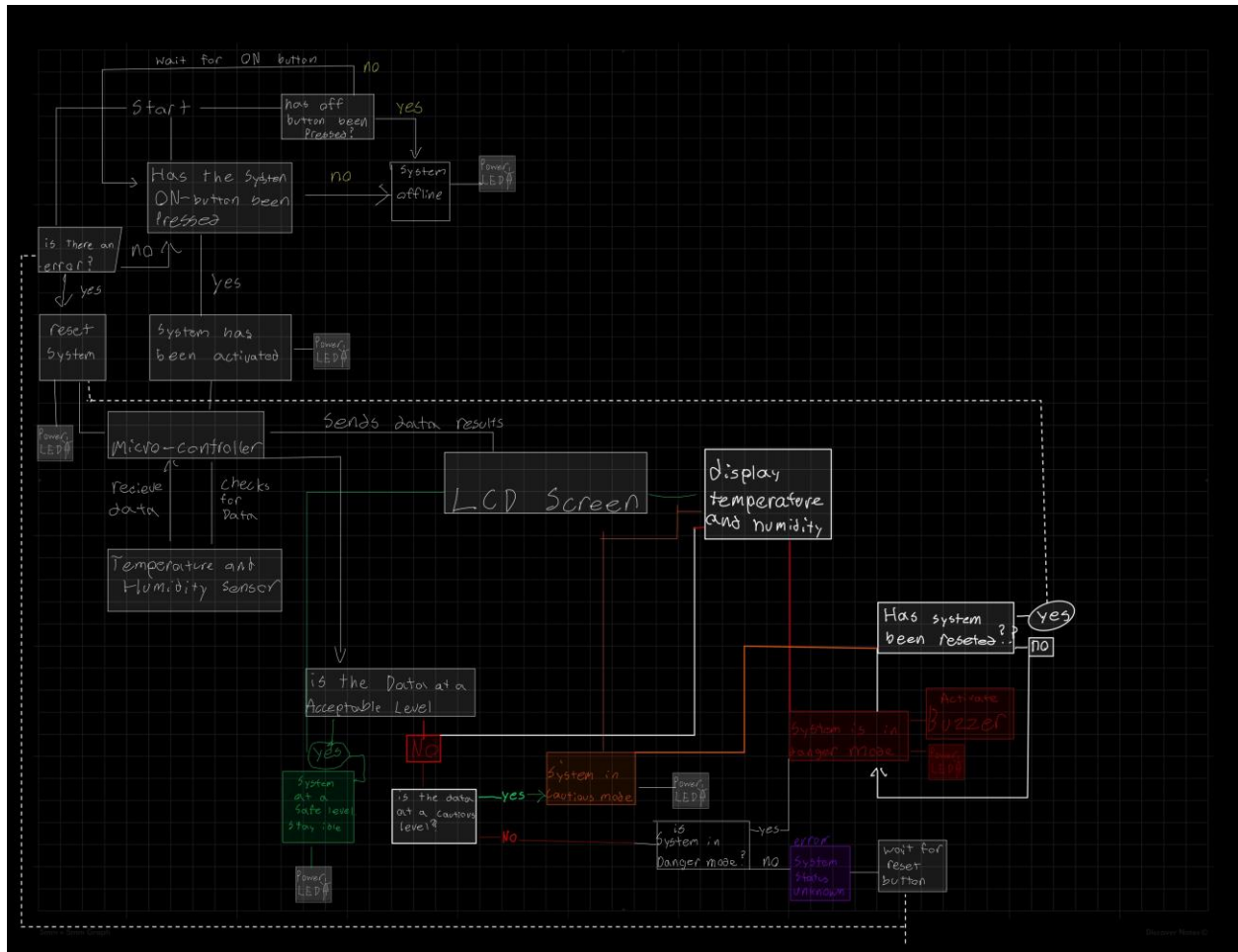
It has three reactions to the temperature and humidity, It will have one of four modes: safe (both temp and humidity are at a safe level and will turn on a green light), danger (one or both are at a dangerous level and will alert the user by flashing red LED and sounding a buzzer), error (turns on a yellow LED and the device doesn't work), and OFF (the device activates a Blue LED and won't read or monitor until turned on)

3.1 Module Design:

There is a current total of 3 Modules

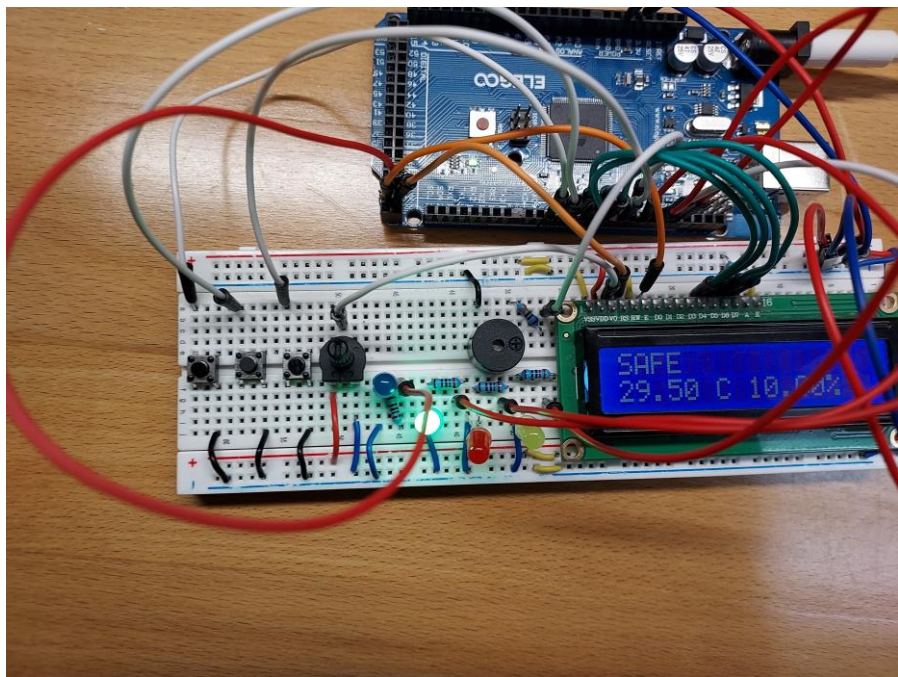
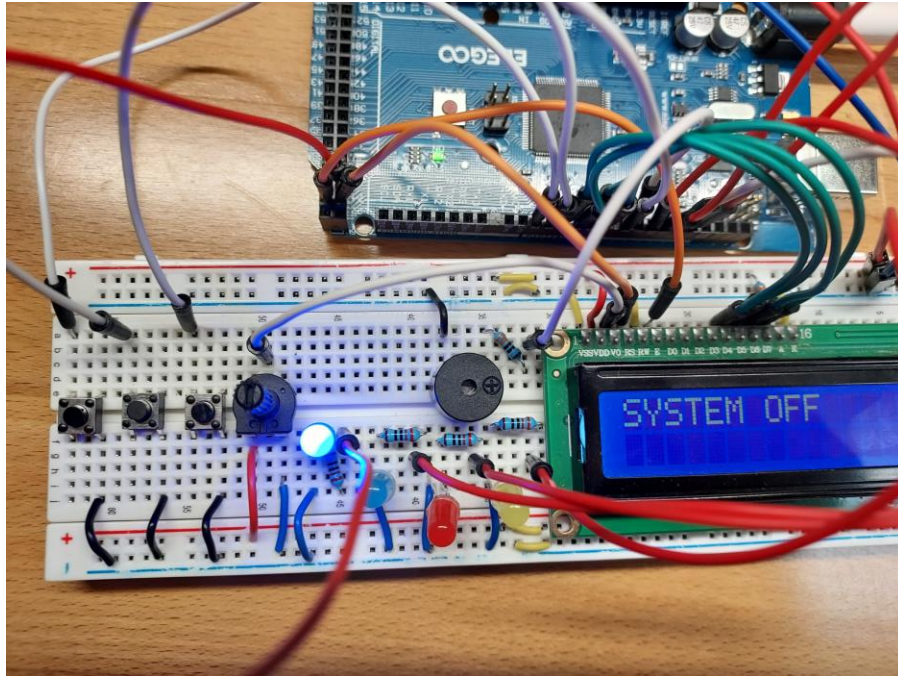
- Temperature & humidity sensor reading Module: Measures the temperature and humidity and sends the information to the LCD display module and control logic module
- LCD Display Module: Receives the data from the Sensor and displays the temperature and humidity
- Control Logic Module: receives data from the Sensor and reacts based on the values, if it senses a value to high or low it will activate the alarm module. If the value is acceptable, it won't activate the alarm module
- Alarm module: activated by the Control logic module and will sound a buzzer and red LED Constantly until reset.

3.2 Control Logic :

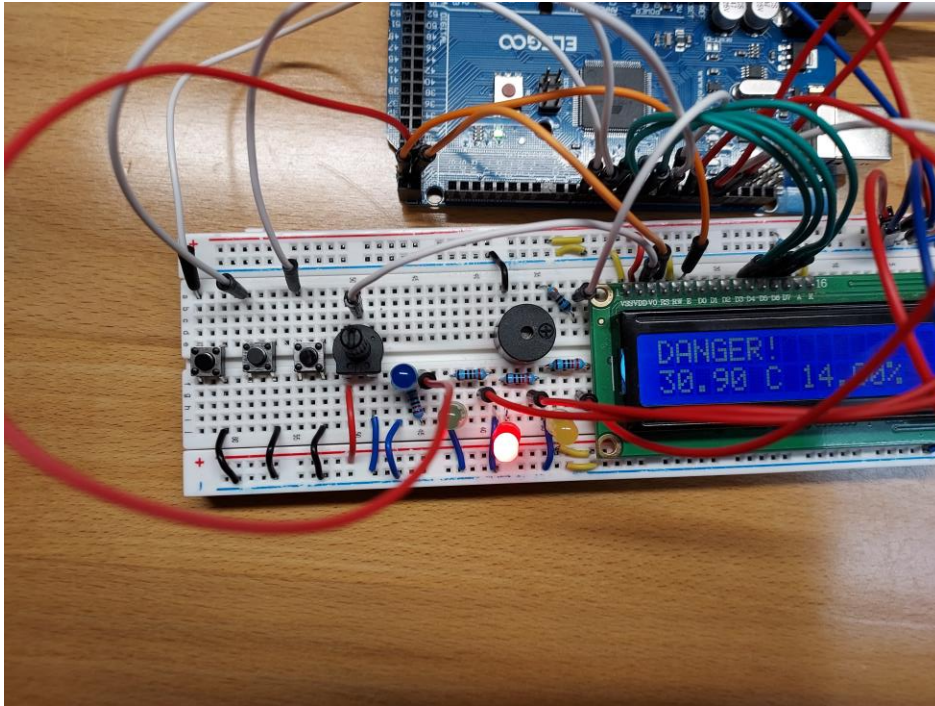


4. RESULT(Circuit images):

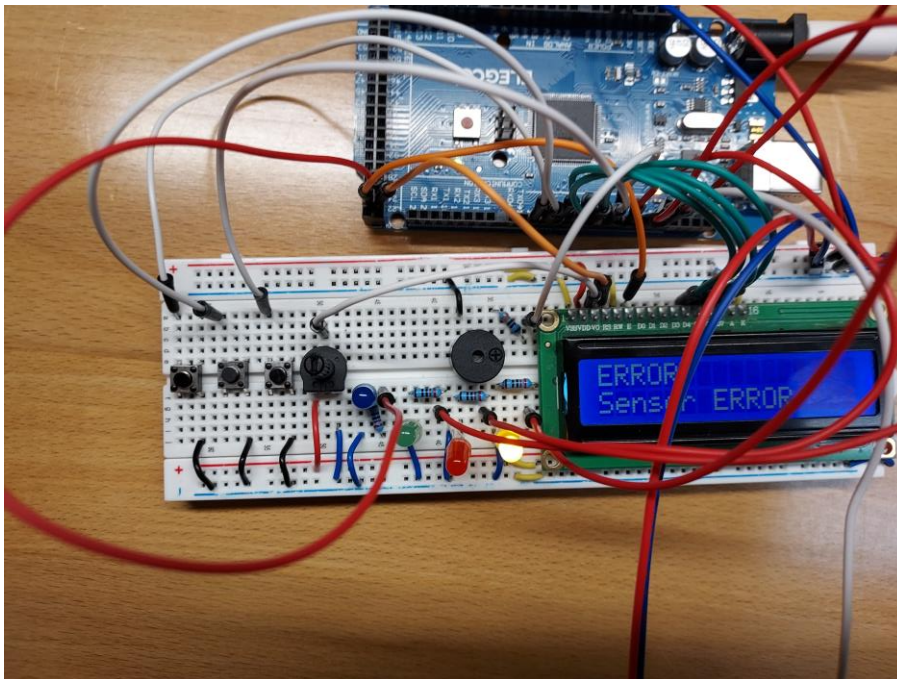
System Offline



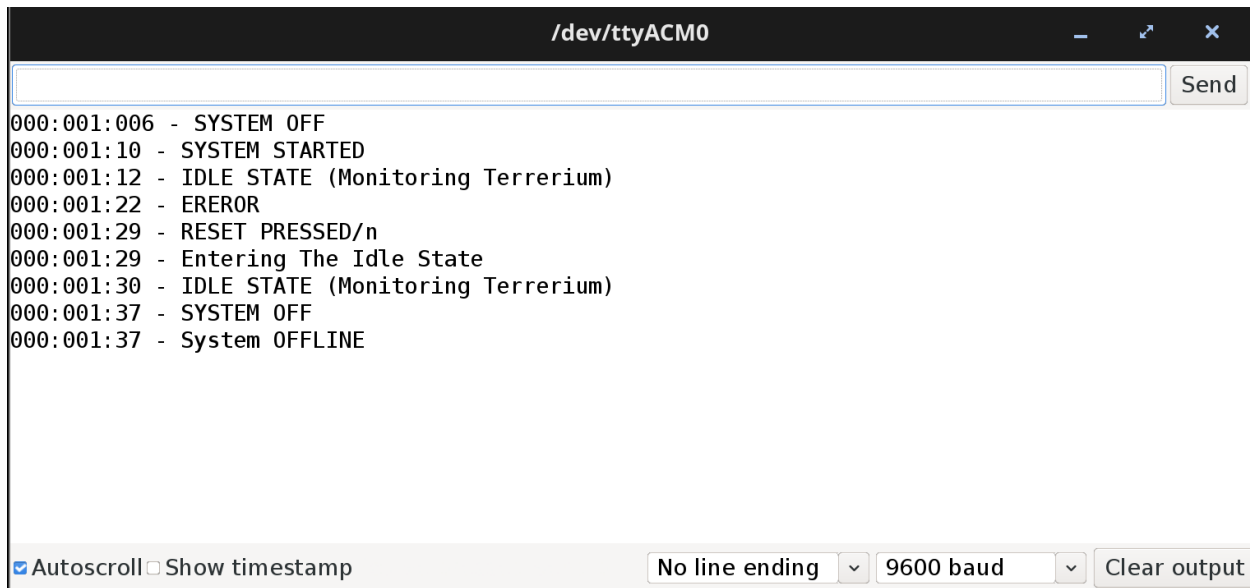
System enters danger mode



System enters the error state



Serial Monitor



```
/dev/ttyACM0
000:001:006 - SYSTEM OFF
000:001:10 - SYSTEM STARTED
000:001:12 - IDLE STATE (Monitoring Terrarium)
000:001:22 - ERROR
000:001:29 - RESET PRESSED/n
000:001:29 - Entering The Idle State
000:001:30 - IDLE STATE (Monitoring Terrarium)
000:001:37 - SYSTEM OFF
000:001:37 - System OFFLINE
```

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5. Environmental & Energy Impact:

The device is built to benefit living creatures and plants. It ensures that whatever is living in the terrarium has a safety net. It is an electrical device and needs to be compact and secured to not only be able to do its job but also that no animal can be harmed by it, such as loose wiring and sharp edges. It's planned to have the device stick to the outside of a terrarium with a wiring connecting to the sensor inside of it. That way it minimizes the device being damaged from intense temperature, humidity, and by any creature within the tank. It will be powered by a 9V1A Adapter. All components will come from the class required kit so that it makes the project more affordable. As for accessibility, besides setting up the device by wiring it, the only thing needed to be done is power the device and place the sensor where you would want it to be. However, this is exclusive for bearded dragons and must be altered to match another animals.

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6. Conclusion:

The Device works. It can be used to monitor the temperature and humidity of a terrarium, the three buttons work, and it constantly checks for any changes and will warn the user of any changes. There are however limitations: It has set levels of humidity and temperature, meaning it will only work to monitor a terrarium for a bearded dragon, Originally it was

planned to have it manually set the temperature and humidity but multiple problems came up like setting the caution reading and danger levels. Time was the biggest hindrance to this project as well as this being a solo project, resulting in less user-friendliness and customization options for other species of reptiles and any other animal where temperature and Humidity must be monitored. It also requires the user to alter the set temperature in the code to adjust it to a different animal. Overall was built for Bearded dragons and that's what it does.